Cosmix New Repository Review and SRS Gap Assessment

Code readiness and reuse decision for the development team

12 October 2026

Recommendation Keep this clone as a candidate foundation, but do not replace the existing system or promise the full SRS in five months on the strength of its current report. The frontend base builds; the backend does not. Many added ERP functions are document storage or status changes, with missing permissions, business handoffs and stock or ledger effects.

The immediate decision is whether this foundation can pass a controlled stabilization and business-flow proof. Adding more repositories before that proof would introduce more integration work without resolving the identified gaps. Three developers are available; the production release scope and team velocity are not yet confirmed.

This assessment combines static source review with actual local build and unit-test checks. Runtime API access, database provisioning, provider integration, browser end-to-end workflows and user acceptance were not verified. Static failure scenarios below are not claims of executed exploits or database incidents.

Review boundaries and source baseline

Reviewed GitHub repository https://github.com/bkhanzaza551-a11y/cosmix_New.git

Pinned commit c4500264431523f7db7960bb3a759a15ca60f08e

Local code C:/Users/Huma.Khan/Downloads/Cosmix_Erp_software/Cosmix_new/code

The original 70-section SRS is the historical reference. The current comparison uses the AI-free detailed SRS v0.4 plus HR Attendance and Employee Mobile SRS v1.2. AI assistants, automatic AI classification, generated replies and predictive forecasting are excluded. Required HR face verification for IN is retained. Rule-based routing and calculations are not excluded AI features.

The 17 presentation groups, 23 panels in the supplied report and 96 detailed specification groups use different levels of grouping. Their counts are not interchangeable completion measures. v0.4 includes proposed contract fields and ownership splits; unresolved business decisions are not treated as approved policy or existing API/schema requirements.

The supplied development report was checked as a set of claims. Its migration commands, proposed radii, SLA hours, OTP length and month-five AI work were not executed or adopted. No tracked application/test code or source Word document was changed. Restore, build and test outputs are local generated files.

Verified build and test results

Check	Observed result	Meaning
Clone and commit	Clone completed and commit pinned	A reproducible code snapshot, not a working delivery.
Backend solution build	FAILED with 16 compile errors	Application layer stops compilation; further errors may be masked.
Domain unit tests	2968 passed and 1 failed of 2969	Compiled domain suite only. Failed permission-count assertion expects 150 but actual count is 174.
Frontend dependency restore	Pinned pnpm 10.28.1 frozen restore passed	Initial bundled newer pnpm mismatch was an environment issue, not counted as a repository defect.
Frontend TypeScript	tsc -b passed	Static typing of existing frontend source.
Frontend production bundle	Vite build passed with warnings	Circular chunks/reexports and large chunks require runtime review; no ERP business acceptance implied.
Frontend unit tests	140 passed in 11 files	Existing unit suite, not the README 154 claim or service/mobile end-to-end coverage.
Application and architecture tests	Not run successfully	Blocked by solution compilation.
Relational migrations and integration tests	Not executed	No database was altered or queried. Current ERP migration gaps are static findings.
External providers and UAT	Not exercised	No Jazz SMS, mail ingestion, WhatsApp, face provider, live GPS or customer acceptance proven.

Commands executed dotnet build src/NOIR.sln; dotnet test tests/NOIR.Domain.UnitTests/NOIR.Domain.UnitTests.csproj --no-build --no-restore; pinned pnpm install --frozen-lockfile --ignore-scripts; node node_modules/typescript/bin/tsc -b; node node_modules/vite/bin/vite.js build; node node_modules/vitest/vitest.mjs run.

The frontend compiler/bundler/test scripts were invoked directly with the bundled Node runtime after pinned restore to avoid the newer bundled package manager automatically reinstalling dependencies. The source and lockfiles remain unchanged. Lifecycle scripts were suppressed during restore.

First release blockers

Backend compile integration

Fourteen observed CS0104 errors are caused by two PaymentMethod enums imported together. Two CS0535 errors arise because CompleteStockTransferCommand does not implement the audit interface members. These are integration defects in the reviewed commit, not caused by an empty customer database. Fixing them would only unlock the next build; it would not complete the missing business workflows.

Code evidence NOIR.Application/GlobalUsings.cs line 155

Code evidence CompleteStockTransfer/CompleteStockTransferCommand.cs line 6

Committed database provisioning

The App migration snapshot and creation chain omit the new ERP aggregates, including BOQ, journals, stock transfers, production, supplier PO, customer PO, invoices, intake, service and QA records. DbSets and EF configurations exist, but are not an applied or reproducibly provisionable SQL schema. Startup runs migrations and seeding automatically, so the application was not started against the user database.

Code evidence Persistence/ApplicationDbContext.cs line 160

Code evidence Persistence/ApplicationDbContextSeeder.cs line 80

The report migration commands also omit --context despite two contexts. The next step is a reviewed migration on an isolated database, after build recovery and schema decisions; not blindly running database update on an existing office database.

Added ERP navigation

Twenty-one advertised added ERP destinations have no corresponding page route. Visible sidebar links fall through to the catch-all redirect. Existing CRM, directory HR and project screens are real; the blanket statement that none of the 23 panels has frontend screens is too broad.

Code evidence portal/Sidebar.tsx line 201

Code evidence src/App.tsx line 233

What is actually in the repository

Source inventory	Count
Tracked files	4512
Tracked C sharp source files including tests and migrations	3349
Tracked frontend TypeScript and TSX files	928
Endpoint source files	72
Endpoint mapping calls in Program	71
HTTP registration patterns in endpoint source	587
Committed App migration files excluding designers and snapshot	28
Backend test source files	966

Counts describe tracked source, not independent APIs, implemented SRS modules, passing cases or a completion percentage. Conditional registrations and multiple routes per file make endpoint and module counts different.

Architecture .NET 10 and React 19 with TypeScript/Vite; SQL Server/EF Core; Wolverine command/query handlers; repository specifications; Finbuckle tenant filtering; Identity/JWT/cookie/API-key authentication; audit infrastructure; Hangfire jobs; SignalR; media/email/webhook support; a generic PWA shell. These are reusable engineering components subject to the observed failures and release checks.

Existing user-facing base User/role/tenant/settings administration, contacts/companies/lead pipeline, employee/department/tag/org-chart directory, project/task boards, ecommerce products/orders/customer directory, commerce receipt/history, media, notifications and commerce reports.

Added ERP behavior BOQ, manual accounts/journals, stock documents and BOM/production have command/query scaffolding with some validations. Procurement, engineering, intake, customer PO, invoicing, service, delivery, helpdesk and QA mainly use direct DbContext endpoints. Direct endpoints are not inherently wrong; the problem is missing equivalent authorization, validation, state and transaction controls.

Definitions An entity is a stored business-record model. An API is a callable server action. A handler is code that processes an action. A migration creates or changes database tables. A UI route connects a URL to a usable screen. A ledger records authoritative stock or financial effects. Idempotency means retrying the same action does not repeat its effect. These layers must agree before a workflow is accepted.

Department and application gap map

The following 17 groups match the presentation structure. SRS identifiers point to current specification groups, not code module IDs. Present means source behavior was found; no row certifies a production-ready complete module.

Sales and commercial

SRS coverage SAL01 to SAL08

Present in code CRM contacts, companies, lead pipeline, activities and customer conversion; customer PO header and comparison skeleton.

Remaining workflow Lead-specific attribution, owned intake handoff, a separately approved and issued quotation/revision, full PO lines and terms comparison, discrepancy resolution and acknowledged Operations handoff.

Code evidence Endpoints/CustomerPoEndpoints.cs line 53

Code evidence WinLead/WinLeadCommandHandler.cs line 41

Estimation

SRS coverage EST01 EST02 and SAL04 SAL05

Present in code BOQ header/items, markup arithmetic, submit/approve/reject and total-only revision records.

Remaining workflow Approved content must be frozen or revised. Full source/equipment/cost versions, alternates, available stock/lead time and approved quotation basis are not completed. Dedicated BOQ workspace is missing.

Code evidence UpdateBoq/UpdateBoqCommandHandler.cs line 38

Engineering

SRS coverage ENG01 to ENG03

Present in code Submittal document/revision counter, direct review action and factor-only unit conversion.

Remaining workflow Technical equipment selection, original/normalized/final values, governed formula versions, AUX definition, submission/review gates and retained approved technical handoff. Celsius/Fahrenheit cannot be represented by factor-only conversion.

Code evidence Engineering/Submittal.cs line 65

Code evidence MasterData/UnitConversion.cs line 39

Operations and field service

SRS coverage OPS01 to OPS08

Present in code Project/task source and UI; manually assigned FieldJob; site coordinate fields and basic start/complete actions.

Remaining workflow Accepted-scope project linkage, qualified and available nearest staff selection, offer/accept/decline/timeout/fallback, authorized assigned-work views, live location evidence and operational zones, safe completion and service closure.

Code evidence Endpoints/FieldJobEndpoints.cs line 81

Code evidence Pm/Project.cs line 8

Inventory and stores

SRS coverage INV01 to INV08

Present in code Commerce product-variant quantities and receipt/history; stock indent/transfer documents; enterprise GRN model.

Remaining workflow Authoritative store/location balances, decimal engineering units, accepted receipt effects, issue/return/adjustment approvals, in-transit transfer and destination discrepancy, source-linked aging and valuation. Complete transfer currently changes document status only.

Code evidence CompleteStockTransfer/CompleteStockTransferCommand.cs line 52

Code evidence Product/ProductVariant.cs line 19

Procurement

SRS coverage PR01 to PR08

Present in code Supplier list/create, PO header/items and approve action; a requisition-like stock indent.

Remaining workflow Demand/BOQ linkage, approved requirement conversion, supplier/agency governance, quotations/comparative selection and reason, value-based authority, issue/acknowledgement/revisions, partial receipt/closure and procurement reporting.

Code evidence Endpoints/ProcurementEndpoints.cs line 69

Code evidence Procurement/PurchaseOrder.cs line 74

Manufacturing

SRS coverage MFG01 MFG02 and OPS02

Present in code BOM validation/scaling and production requirement snapshot, release/start/progress/material-issue/complete handlers.

Remaining workflow Approved project demand and frozen revisions, material stock debit and returns, lifecycle-safe issue, once-only concurrent progress, applicable QA/rework, finished-goods receipt and controlled handoff. Quantity counters alone do not move stock.

Code evidence IssueMaterial/IssueMaterialCommandHandler.cs line 50

Code evidence Manufacturing/ProductionOrder.cs line 182

Quality assurance

SRS coverage QA01 QA02

Present in code Inspection record with stage, text checklist, result, defects/actions and evidence URL.

Remaining workflow Versioned approved criteria, trusted reviewer/evidence, correction and reinspection, linkage to exact source version, applicable receiving/production/project release gates. QA should be applied at agreed checkpoints, not forced onto every transaction.

Code evidence Qa/QaInspection.cs line 67

Human resources

SRS coverage HR01 to HR08 and HR v1.2

Present in code Meaningful employee/department/tag directory, manager checks, user link, import/export and headcount reports.

Remaining workflow Expanded private employee fields, controlled activation/offboarding, dated assignments/shifts/holidays, face IN and no-face OUT, raw punches/sessions, overnight/offline/time rules, corrections/month locks, leave/OT, meal/fuel/vehicle/loan effects and locked payroll-input handoff. No attendance engine was found.

Code evidence Hr/Employee.cs line 8

Code evidence Persistence/ApplicationDbContext.cs line 135

Office employee mobile panel

SRS coverage EMP01 to EMP06 EMP08

Present in code Generic signed-in account profile/sessions and shared notifications; responsive admin HR screens.

Remaining workflow Owned employee profile/document requests, IN/OUT/calendar, correction/site reports, leave/OT, salary effects/claims, and current-team supervisor decisions. A PWA manifest or account profile is not this domain app.

Code evidence src/App.tsx line 215

Code evidence Endpoints/AuthEndpoints.cs line 352

Field and site employee panel

SRS coverage EMP02 EMP03 EMP06 EMP07 EMP08

Present in code FieldJob server record and manual status API, but no rendered technician destination.

Remaining workflow Assigned jobs, acceptance/decline, safe job details, duty/site presence and face-IN attendance distinction, evidence/photos/material reporting, sync/retry, trusted recipient proof and appropriate supervisor views.

Code evidence portal/Sidebar.tsx line 232

Code evidence Endpoints/FieldJobEndpoints.cs line 36

Customer application

SRS coverage CUS01 to CUS07

Present in code Customer UserId association and administrative directory; complaint list/create/assign/resolve source.

Remaining workflow Customer-owned complaint/site/evidence submission, receipt/timeline, visit/rescheduling communication, confirm/dispute/authorized no-response closure and own service history. CustomerId from the request is not proof of ownership.

Code evidence Customer/Customer.cs line 15

Code evidence Endpoints/ComplaintEndpoints.cs line 36

Accounts

SRS coverage ACC01 to ACC08

Present in code Chart of accounts and manual journal CQRS; invoice/payment models and direct actions.

Remaining workflow Trusted source/version references, vendor bills/AP, eligible accounts/open periods, exact approved balance/rounding rules, journal approval/reversal treatment, authoritative trial balance and AR/AP aging, closing, HR package verification. Invoice/payment events do not create linked journal entries.

Code evidence Accounting/JournalEntry.cs line 118

Code evidence Endpoints/InvoicingEndpoints.cs line 73

Finance

SRS coverage FIN01 to FIN08

Present in code Executive invoice-total KPI and inherited commerce reports.

Remaining workflow Financial oversight must derive from the single Accounts ledger, with approved margin/tax/currency/rate/value authority and period/HR decisions. Current revenue sum mixes currencies and drops PartiallyPaid invoices. Accounts/Finance ownership requires sign-off; a second ledger is not recommended.

Code evidence Endpoints/ExecutiveDashboardEndpoints.cs line 21

Administration

SRS coverage ADM01 to ADM08

Present in code User/role/settings administration and platform tools; active Ticket/comment storage; project tasks.

Remaining workflow Scoped access-change approval, common reference/approval ownership, trusted helpdesk identity/status/SLA, special task/event workflow and governed imports/reminders. User-supplied ticket/comment author is not trustworthy. A richer unused HelpdeskTicket model must not be counted as API behavior.

Code evidence Endpoints/TicketEndpoints.cs line 36

Code evidence Persistence/ApplicationDbContext.cs line 191

Superadmin and platform administration

SRS coverage SA01 to SA08

Present in code Platform-admin and tenant-admin roles, permission policy infrastructure, feature definitions, audit/configuration and background jobs.

Remaining workflow Complete permission registration/enforcement across added APIs, secure bootstrap/secrets, controlled deployment/backup/restore, integration failure ownership and verified recovery. Technical admin is not automatic CEO, payroll or domain approval authority.

Code evidence Common/Permissions.cs line 535

Code evidence NOIR.Web/Program.cs line 436

Shared functions and management visibility

SRS coverage SH01 to SH10 and common contracts

Present in code Media, notifications, audit, email templates, outbound webhooks, background-job infrastructure and commerce reports.

Remaining workflow One governed master/calculation/approval contract, reliable inbound/outbound provider reconciliation, safe files/export scope, immutable business versions, transaction/retry protection, operational location control and traceable ERP/CEO reports. Infrastructure does not supply the missing business engines.

Code evidence Endpoints/IntakeEndpoints.cs line 26

Code evidence NOIR.Web/Program.cs line 814

Shared controls required across departments

Shared function	Existing foundation and remaining control
SH01 Master data	Product/category/brand/attribute basics exist. Store, approved tax/rates, controlled units and effective-date versions are not complete.
SH02 Calculations	Local quantity-rate/markup/exclusive-tax arithmetic exists. Shared approved tax, WHT, currency, costing and historical inputs are missing.
SH03 Approvals	Some BOQ/journal/stock/production action permissions exist. No common business authority/value-limit/current-assignment decision contract across added APIs.
SH04 Communication	Email templates and outbound webhook infrastructure exist. Email/WhatsApp/social/SMS intake connectors, durable source IDs, verified callbacks, unknown-outcome reconciliation and controlled replies are missing.
SH05 Files and exports	Generic media/file and directory export tools exist. Business-owned evidence/version linkage and HR/customer/private-field access must be enforced for each workflow.
SH06 Tasks and notices	Project tasks and notifications exist. Assigned service acceptance, special task ownership and required insurance/approval/SLA reminder policies are not complete.
SH07 Reports and KPIs	Commerce/headcount and project dashboards exist. Authoritative ERP aging/trial balance/payroll/service and drill-down lifecycle metrics are incomplete.
SH08 History and audit	HTTP/handler/entity logging exists. It does not freeze approved documents, authenticate a request-supplied reviewer, or make retries once-only.

Shared function	Existing foundation and remaining control
SH09 Location and zones	Site coordinate fields exist. Fresh staff presence, eligibility/distance routing, reviewed zone events and separate attendance geofence logic are missing.
SH10 Platform and recovery	Identity/tenancy/configuration/job infrastructure exists. Runtime authorization, fresh SQL provisioning, hardening, deployment, backup restore and release evidence remain unverified.

Important boundary Tenant filtering was found and is not dismissed. It isolates modeled tenant data; it does not prove that a signed-in person is the assigned technician, owning customer, current supervisor or authorized financial approver. Dedicated negative role/ownership and tenant-header/JWT mismatch tests remain necessary.

Detailed code findings

P1 identifies a blocker or integrity/access risk for the affected selected workflow. P2 identifies a material correctness or integration gap. Findings below are static observations except for the separately documented build/test results. Repeated concerns across departments are not separate effort estimates. Absence means not found in the reviewed tracked implementation; an external system was not assumed to fill the gap.

Sales engineering projects and UI

F01 P1 Added menu entries do not open ERP screens

The 21 destinations above have no page routes; selecting a visible link falls through to the / redirect. No BOQ builder, PO comparison, engineering review, intake inbox or inspection workspace is wired. Passing TypeScript/Vite compilation does not detect absent router declarations.

Code evidence portal/Sidebar.tsx line 201

Code evidence portal/Sidebar.tsx line 270

Code evidence portal/Sidebar.tsx line 298

F02 P1 New review APIs authenticate users but do not authorize business decisions

Intake, engineering, customer PO, QA and unit-conversion groups use bare RequireAuthorization() without scoped permission or feature checks. Engineering accepts ReviewedBy from the request and QA accepts InspectedBy; neither derives the decision actor from current-user context. A signed-in caller can attempt these mutations without a domain-approver policy, including when the module is disabled, and can supply another reviewer's name.

Code evidence Endpoints/IntakeEndpoints.cs line 9

Code evidence Endpoints/EngineeringEndpoints.cs line 9

Code evidence Endpoints/EngineeringEndpoints.cs line 48

F03 P1 Approval does not freeze BOQ content or provide a controlled revision route

Header updates and item add/update/delete load the BOQ by ID and save without checking its status. The update specification filters by ID only. Thus an approved BOQ can have its quantities/rates/markup changed while still reporting Approved.

Code evidence UpdateBoq/UpdateBoqCommandHandler.cs line 38

Code evidence AddBoqItem/AddBoqItemCommandHandler.cs line 38

Code evidence UpdateBoqItem/UpdateBoqItemCommandHandler.cs line 45

F04 P1 Customer PO reconciliation can claim an exact match without comparing items

The endpoint always supplies an empty PO-item list. The engine loops only over PO items and declares success when its discrepancy list is empty. Equal currency and total therefore produce an Exact Match even when the actual customer PO has different or omitted items; its success reason says all items match.

Code evidence Endpoints/CustomerPoEndpoints.cs line 53

Code evidence Commercial/ExceptionRecord.cs line 70

Code evidence Commercial/ExceptionRecord.cs line 99

F05 P1 The commercial quotation and accepted scope handoff are missing

The commercial implementation contains a customer-PO header, discrepancies and a BOQ-based comparison. A separate issued quotation/revision lifecycle, margin/discount approvals, customer response, commercial hub query and accept/reject/receiving-owner handoff endpoints were not found. PO Accept/Reject methods exist only on the entity.

Code evidence Endpoints/CustomerPoEndpoints.cs line 23

Code evidence Endpoints/CustomerPoEndpoints.cs line 45

Code evidence Commercial/CustomerPurchaseOrder.cs line 78

F06 P1 QA records outcomes but does not enforce quality release

QA has a stage enum and text checklist, not versioned, approved checkpoint criteria. RecordResult overwrites result, reviewer, defect/action/evidence fields without a valid-transition, required-evidence or reinspection check. References are optional ID/type fields; creation does not load the source or bind its approved version.

Code evidence Qa/QaInspection.cs line 32

Code evidence Qa/QaInspection.cs line 67

Code evidence Endpoints/QaEndpoints.cs line 39

F07 P1 Projects are standalone task workspaces not controlled ERP execution records

Project fields and creation requests have no customer, accepted PO, quotation or approved technical-version link. Creation starts Active and seeds task columns, independent of commercial acceptance. Completion permits Active to Completed without evaluating outstanding tasks, material dependencies, delivery evidence or applicable QA.

Code evidence Pm/Project.cs line 8

Code evidence Pm/Project.cs line 63

Code evidence DTOs/ProjectDtos.cs line 89

F08 P2 Intake is manual record storage not a multichannel communication workflow

Channel enum values and an authenticated create endpoint do not implement email/WhatsApp/SMS/social adapters. Searches found no IMAP/POP intake worker, provider event receipt, callback reconciliation or caller creating IntakeMessage outside this endpoint. The model lacks provider account/event/thread keys, retained attachment links, confirmed customer/site/business links and receiving-owner acknowledgement/history. Assignment can simply overwrite department/user/status.

Code evidence Endpoints/IntakeEndpoints.cs line 26

Code evidence Intake/IntakeMessage.cs line 32

Code evidence Intake/IntakeMessage.cs line 71

F09 P2 Engineering review skips submission and retained technical versions

Review accepts any valid SubmittalStatus directly, including approving a newly created Draft, with optional comments. Submit and IncrementRevision are entity methods with no exposed actions/callers found. IncrementRevision changes the document URL and status on the same record, retaining no prior document revision; it does not clear prior reviewer/time/comments.

Code evidence Endpoints/EngineeringEndpoints.cs line 43

Code evidence Engineering/Submittal.cs line 59

Code evidence Engineering/Submittal.cs line 65

F10 P2 Conversion master cannot represent approved formula effective date rules

UnitConversion holds a factor, category and always-active initial state, without approver/effective dates/version. Convert computes only value times factor; it cannot represent an affine conversion such as Celsius to Fahrenheit, which the entity comment and original SRS list as examples. Only list/create endpoints exist, with no governed activation/review or conversion-result provenance.

Code evidence MasterData/UnitConversion.cs line 5

Code evidence MasterData/UnitConversion.cs line 13

Code evidence MasterData/UnitConversion.cs line 35

F11 P2 Reference validation retries and conflict handling are incomplete

The direct intake/engineering/PO/QA handlers create records from request IDs/strings without loading and validating referenced customer/project/BOQ/department/user/source records. Request models have no field validators in these paths. PO creation permits a supplied amount without a positive-value guard in its factory; QA permits missing reference/evidence. Database length/unique indexes are not field-level business validation.

Code evidence Endpoints/IntakeEndpoints.cs line 28

Code evidence Endpoints/EngineeringEndpoints.cs line 28

Code evidence Endpoints/CustomerPoEndpoints.cs line 25

F12 P2 CRM lacks full lead attribution and permits an inconsistent pipeline stage

Attribution is a ContactSource enum; the UI offers Web/Referral/Social/Cold/Event/Other. Lead has no inquiry/source/campaign/referrer/keyword/landing-page/city fields. Source attribution cannot remain specific to each lead when a contact has several inquiries. MoveLeadStage also fails to check that the target stage belongs to the lead pipeline. A direct request can change StageId to another pipeline while preserving the old PipelineId, leaving board and history inconsistent.

Code evidence Crm/Lead.cs line 8

Code evidence MoveLeadStage/MoveLeadStageCommandHandler.cs line 48

Code evidence MoveLeadStage/MoveLeadStageCommandHandler.cs line 71

Finance supply and manufacturing

P1 D01 Procurement GRN and invoicing consequential APIs enforce authentication only

An authenticated tenant user can reach supplier creation, PO creation and approval, GRN creation, and invoice/payment actions without the corresponding business permission or module gate. PO approval has no trusted reviewer identity, amount limit or decision evidence. Tenant filtering does not supply these within-tenant authority checks.

Code evidence Endpoints/JournalEntryEndpoints.cs line 141

Code evidence Endpoints/ProcurementEndpoints.cs line 43

Code evidence Endpoints/ProcurementEndpoints.cs line 94

P1 D02 Receipt confirmation can commit partial stock while returning failure

Trigger: a StockOut receipt has two items; the first has sufficient stock, the second does not. Confirmation changes the header first, then the first movement calls a logger which saves the shared DbContext; the second returns a validation failure without rollback. The database can retain a Confirmed header, first deduction and first movement while omitting the second deduction. Retry is then blocked by the Confirmed state.

Code evidence ConfirmInventoryReceipt/ConfirmInventoryReceiptCommandHandler.cs line 48

Code evidence ConfirmInventoryReceipt/ConfirmInventoryReceiptCommandHandler.cs line 79

Code evidence ConfirmInventoryReceipt/ConfirmInventoryReceiptCommandHandler.cs line 95

P1 D03 Missing receipt products variants produce false successful confirmation

Creation validates references, but a product can become unavailable before confirmation. The confirmation handler uses continue when the product or variant is missing, then saves the Confirmed receipt and returns success. A missing sole item produces zero stock effects with a successful confirmation.

Code evidence ConfirmInventoryReceipt/ConfirmInventoryReceiptCommandHandler.cs line 62

Code evidence ConfirmInventoryReceipt/ConfirmInventoryReceiptCommandHandler.cs line 65

Code evidence ConfirmInventoryReceipt/ConfirmInventoryReceiptCommandHandler.cs line 106

P1 D04 Completed stock transfers execute no stock transfer

An Approved transfer goes straight to Completed; the handler only calls Complete, saves and broadcasts the document. No source debit, destination credit, availability check, reservation, dispatch/in-transit or actual receiving discrepancy exists.

Code evidence CompleteStockTransfer/CompleteStockTransferCommand.cs line 40

Code evidence CompleteStockTransfer/CompleteStockTransferCommand.cs line 52

Code evidence Stock/StockTransfer.cs line 152

P1 D05 Manufacturing issue completion records do not consume or receive stock

IssueMaterial only increments ProductionOrderItem.IssuedQuantity. CompleteProductionOrder only sets status/audit after positive ProducedQuantity. Thus Release > Start > Progress(1) > Complete can finish with zero material issues, and a material issue does not check or deduct any store/variant balance. Finished goods are not received into inventory.

Code evidence IssueMaterial/IssueMaterialCommandHandler.cs line 50

Code evidence CompleteProductionOrder/CompleteProductionOrderCommandHandler.cs line 47

Code evidence Manufacturing/ProductionOrder.cs line 182

P1 D06 Material issue bypasses order lifecycle and can invalidate requirement quantities

IssueMaterial loads the order/item without checking Draft, Released, InProgress, Completed or Cancelled status. A Draft or already Cancelled/Completed order can receive additional issues while the quantity cap permits it. On a Draft order, issue 8 against required 10, then reduce planned quantity so the new requirement is 5: UpdateProductionOrder updates the existing requirement without checking IssuedQuantity, leaving issued 8 > required 5.

Code evidence IssueMaterial/IssueMaterialCommandHandler.cs line 32

Code evidence Manufacturing/ProductionOrderItem.cs line 69

Code evidence UpdateProductionOrder/UpdateProductionOrderCommandHandler.cs line 83

P1 D07 Additive production and stock operations have no request deduplication

Resending the same progress request while InProgress adds produced/scrap quantities again; the request/handler supplies no stable operation key or prior-result lookup. Material issue likewise increments again until its cap, and manual stock movement repeats a quantity delta.

Code evidence Manufacturing/ProductionOrder.cs line 163

Code evidence ReportProductionProgress/ReportProductionProgressCommandHandler.cs line 43

Code evidence Manufacturing/ProductionOrderItem.cs line 78

P1 D08 Concurrent journal edits can bypass the Draft only posting invariant

Two requests can load the same Draft entry. One posts its balanced lines; the other subsequently inserts an additional debit line based on its earlier Draft check. Neither a version precondition nor parent concurrency token makes the line edit fail after posting, so the posted entry can become unbalanced.

Code evidence AddJournalLine/AddJournalLineCommandHandler.cs line 42

Code evidence AddJournalLine/AddJournalLineCommandHandler.cs line 68

Code evidence PostJournalEntry/PostJournalEntryCommandHandler.cs line 42

P1 D09 Journal posting permits imbalance and omits account period eligibility checks

Debit 1,000.00 and credit 999.99 are accepted because both creation and Post reject only differences greater than 0.01. Arbitrary decimal precision is accepted before persistence rounds lines to two places. Post also checks no fiscal period or current account eligibility; line creation verifies existence but not IsActive.

Code evidence CreateJournalEntry/CreateJournalEntryCommandHandler.cs line 41

Code evidence Accounting/JournalEntry.cs line 118

Code evidence Configurations/JournalLineConfiguration.cs line 22

P1 D10 PO GRN creation accepts invalid references and contradictory quantities

Create a PO with a nonexistent SupplierId/ProductId, negative quantity/rate, or no items: the direct API calls factories without business validation; supplier/product are scalar IDs with no configured FK in the shown mappings. Approve then unconditionally sets Approved, with no source-state or completeness guard. GRN can store received 10, accepted 20, rejected -1, without a PO-item reference or quantity reconciliation.

Code evidence Endpoints/ProcurementEndpoints.cs line 69

Code evidence Endpoints/ProcurementEndpoints.cs line 79

Code evidence Procurement/PurchaseOrder.cs line 74

P1 D11 Invoice payment actions can corrupt settlement state

Record payment against a Draft invoice, use an unrelated CustomerId, submit negative/zero amounts, overpay, or repeat a transaction under a different PaymentNumber: no handler/domain guard rejects these cases. RecordPayment directly increments PaidAmount, allowing negative outstanding or inconsistent status. Issue unconditionally sets Issued, including a previously Paid invoice. Invoice lines accept negative quantities/rates/tax without business validation.

Code evidence Endpoints/InvoicingEndpoints.cs line 73

Code evidence Invoicing/PaymentRecord.cs line 21

Code evidence Invoicing/SalesInvoice.cs line 63

P1 D12 Manual stock changes and their history can commit separately

CreateStockMovement first saves the stock delta, then invokes the logger that separately saves the movement. If logging fails, stock has already committed and the API fails; a retry can apply another delta without a complete audit trail.

Code evidence CreateStockMovement/CreateStockMovementCommandHandler.cs line 94

Code evidence Inventory/InventoryMovementLogger.cs line 52

Code evidence CreateStockMovement/CreateStockMovementCommandHandler.cs line 110

P2 D13 ERP sidebar destinations lead to the catch all redirect

The sidebar advertises accounting, stock, manufacturing, procurement, GRN and invoicing destinations, but App.tsx registers none of their corresponding pages. Opening them reaches the catch-all redirect to /.

Code evidence portal/Sidebar.tsx line 208

Code evidence portal/Sidebar.tsx line 216

Code evidence portal/Sidebar.tsx line 224

P2 D14 Stock documents can pass workflow with invalid content

Create a one-item indent/transfer, remove its last item, then submit/approve: entity transition guards inspect status only, not item count. Transfers can ultimately complete empty. Locations can also become identical after normalization: input "Store A" and " Store A " passes raw string comparison but both are trimmed to "Store A". Update checks have the same normalization ordering.

Code evidence Stock/StockIndent.cs line 74

Code evidence Stock/StockIndent.cs line 99

Code evidence Stock/StockTransfer.cs line 74

P2 D15 The chart of accounts hierarchy permits ancestor cycles

Create A, create B with parent A, then set A's parent to B. UpdateAccount rejects only direct self-parenting and verifies the parent exists; both conditions pass. The self-FK does not reject a cycle.

Code evidence UpdateAccount/UpdateAccountCommandHandler.cs line 51

Code evidence UpdateAccount/UpdateAccountCommandHandler.cs line 59

Code evidence Configurations/AccountConfiguration.cs line 34

P2 D16 Executive revenue is currency unsafe and changes incorrectly on partial payment

The executive KPI sums native invoice totals without currency grouping/conversion and includes only Paid or Issued. Recording a small partial payment changes an invoice to PartiallyPaid and removes its entire total from this KPI.

Code evidence Endpoints/ExecutiveDashboardEndpoints.cs line 21

Code evidence Invoicing/SalesInvoice.cs line 75

Code evidence Invoicing/SalesInvoice.cs line 21

P2 D17 Duplicate BOM components pass validation and hit a database uniqueness error

Submit two BOM component lines with the same ProductId, or add the same component again. The factory creates fresh line IDs and AddItem checks duplicate line Id only. The mapping requires unique (BillOfMaterialId, ProductId), so the request reaches SaveChanges with invalid data rather than a clear duplicate-component error.

Code evidence CreateBillOfMaterial/CreateBillOfMaterialCommandHandler.cs line 69

Code evidence CreateBillOfMaterial/CreateBillOfMaterialCommandValidator.cs line 35

Code evidence Manufacturing/BillOfMaterial.cs line 79

People mobile and service

PS 01 P1 New service APIs omit business permissions and ownership

A valid signed-in tenant user can enter job creation/assignment/start/completion, complaint assignment/resolution, ticket status changes and delivery confirmation without a named service permission. Lists and GUID detail lookups also omit customer/employee ownership. This permits unrelated users to read or alter other same-tenant service records if the endpoints are made runnable. Authentication-only API keys also do not reach the permission-scope handler.

Code evidence Endpoints/FieldJobEndpoints.cs line 15

Code evidence Endpoints/FieldJobEndpoints.cs line 94

Code evidence Endpoints/ComplaintEndpoints.cs line 89

PS 02 P1 Customer OTP is caller chosen disclosed optional and reusable

Creation accepts CustomerOtp, copies it to a public entity property, and returns the entity. GET also returns entities containing the plaintext value. An omitted/empty stored OTP disables verification entirely; a nonempty value has no expiry, failed-attempt budget or consumed state. Repeating confirmation can reuse it. This does not prove recipient identity or a customer-controlled SMS response.

Code evidence Endpoints/FieldJobEndpoints.cs line 70

Code evidence Endpoints/FieldJobEndpoints.cs line 76

Code evidence Fsm/FieldJob.cs line 27

PS 03 P1 Jobs and deliveries accept invalid transitions and rewrite completion evidence

StartJob() unconditionally sets InProgress and StartedAt, including after completion; CompleteJob() can complete an unstarted/unassigned job and resets CompletedAt on each call. ConfirmDelivery() can move a newly created Draft straight to Delivered, without dispatched state, OTP when absent, or proof; retry changes delivery time and can replace proof.

Code evidence Fsm/FieldJob.cs line 77

Code evidence Fsm/FieldJob.cs line 83

Code evidence Endpoints/FieldJobEndpoints.cs line 107

PS 04 P1 Helpdesk requester and comment identity can be forged

The API trusts CreateTicketRequest.UserId and AddTicketCommentRequest.UserId, passes them directly into the domain, and never derives them from authenticated identity. A signed-in caller can attribute a ticket/comment to a different or nonexistent user. Generic audit actor and the displayed author can therefore disagree.

Code evidence Endpoints/TicketEndpoints.cs line 36

Code evidence Endpoints/TicketEndpoints.cs line 49

Code evidence Endpoints/TicketEndpoints.cs line 73

PS 05 P1 Employee activation bypasses the HR v1 2 lifecycle

The factory creates Active employees immediately; the create handler validates directory fields then persists without Draft, submission, required documents, face enrollment or independent approval. The status enum has no Draft/Pending Approval/Retired, and its Suspended value has no status-change route. Base behavior is intentional in existing tests, but conflicts with the newer HR SRS.

Code evidence Hr/Employee.cs line 70

Code evidence Enums/EmployeeStatus.cs line 6

Code evidence CreateEmployee/CreateEmployeeCommandHandler.cs line 80

PS 06 P1 HR termination does not disable or revoke the linked account

Deactivation updates employee status/end date and reporting links, then saves. It neither disables the linked ApplicationUser nor revokes sessions; the deactivation event has a webhook mapping but no identified local account-revocation handler. Login checks the user account's IsActive rather than the linked employee's status. A terminated employee whose user account remains active can still authenticate.

Code evidence DeactivateEmployee/DeactivateEmployeeCommandHandler.cs line 37

Code evidence Hr/Employee.cs line 157

Code evidence Webhooks/WebhookEventTypeRegistry.cs line 97

PS 07 P2 Delivery quantities and service references lack business validation

Challan creation can contain no items or nonpositive quantities; AddItem/constructor simply copy input, and quantity configuration sets precision only. FSM/complaint assignment accepts any technician GUID without checking active employee, skill, certification, site or availability. Customer/project/product references are copied without scoped existence checks in these paths.

Code evidence Endpoints/FulfillmentEndpoints.cs line 42

Code evidence Fulfillment/DeliveryChallanItem.cs line 19

Code evidence Configurations/FulfillmentAndStoresConfigurations.cs line 35

PS 08 P2 Service module toggles are not enforced on their APIs

FSM, complaints, helpdesk and fulfillment are declared toggleable, yet their endpoint groups omit RequireFeature and directly execute DbContext operations instead of feature-gated message handlers. Disabling a module therefore has no identified server gate on these routes.

Code evidence Erp/FsmModuleDefinition.cs line 10

Code evidence Erp/ComplaintsModuleDefinition.cs line 10

Code evidence Erp/HelpdeskModuleDefinition.cs line 10

PS 09 P2 Service navigation has no rendered destination

Sidebar links target FSM jobs, complaints, challans and helpdesk tickets, but App.tsx declares none of those routes and redirects unmatched paths to /. No service API client/page wiring was found.

Code evidence portal/Sidebar.tsx line 232

Code evidence portal/Sidebar.tsx line 239

Code evidence portal/Sidebar.tsx line 254

PS 10 P2 Support state changes lose the meaning of closure

Ticket status can move freely from Closed back to Open/InProgress, while ClosedAt remains set; closing again overwrites it. There is no resolution evidence or controlled reopen decision. Complaint assignment similarly overwrites Resolved with Assigned while retaining prior ResolutionNotes/ResolvedAt; resolve can also run from any state.

Code evidence Helpdesk/Ticket.cs line 69

Code evidence Endpoints/TicketEndpoints.cs line 61

Code evidence Complaints/CustomerComplaint.cs line 53

PS 11 P2 Expected OTP rejection becomes a server error

Wrong nonempty OTP throws InvalidOperationException from FieldJob/DeliveryChallan. Endpoints do not map it to a validation result, and the global exception switch sends unrecognized exceptions to the 500 handler.

Code evidence Fsm/FieldJob.cs line 85

Code evidence Fulfillment/DeliveryChallan.cs line 75

Code evidence Middleware/ExceptionHandlingMiddleware.cs line 70

Business flow acceptance gaps

Inquiry to accepted execution

Retain source/customer/thread -> reviewed intake owner -> validated and versioned BOQ/technical selection -> approved and issued quotation -> complete customer PO comparison -> authorized discrepancy resolution -> acknowledged Operations handoff -> linked execution project. CRM win and matching total do not establish this chain.

Supply to usable stock

Approved demand -> approved supplier selection/PO -> expected versus actual receipt -> applicable QA/rejection disposition -> once-only accepted store movement. Receipt header confirmation without the stock effect is not success; multiple item effects must be atomic.

Transfer and production

Source availability -> authorized dispatch/in-transit -> actual destination receipt/discrepancy -> conserved balances. Production must tie approved demand and frozen requirements to material issues/returns, progress, applicable QA and accepted finished goods.

Complaint to customer closure

Owned customer complaint/site -> eligible available qualified staff before distance ranking -> offer acceptance or next-candidate/supervisor fallback -> assigned work/evidence -> customer confirm/dispute or authorized no-response review -> controlled closure/reopen. A technician Resolve setter is not customer acceptance.

Employee attendance to Accounts input

Approved activation/enrollment -> live one-to-one face IN proof -> immutable IN/OUT events with no face for OUT -> sessions/date-effective policy -> separate corrections/leave/OT/approved effects -> reviewed monthly lock -> versioned Accounts handoff. Operational zone warnings remain separate from automatic payroll/discipline.

Financial source to reports

Approved billing/receipt/bill/journal treatment -> eligible accounts and open period -> one authoritative linked posting path -> reconciled trial balance/AR/AP views -> authorized closing/correction. Payment approval/settlement and full payroll/bank transfer are only included when management approves that scope.

Every accepted route needs authorized and denied API tests, invalid references, legal/illegal transitions, failed-provider outcomes, duplicate/stale requests, concurrent edits and retained evidence. Add actual web/mobile end-to-end tests against a migrated isolated SQL database; inherited unit-test totals cannot substitute.

Corrections to the supplied development report

Report claim or proposal	Assessment
Backend 45 to 50 percent and overall 35 to 40 percent	Not validated. No signed requirement weights, acceptance denominator, runtime evidence or current HR/mobile coverage supports these percentages.
All 23 panel schemas and APIs completed	Models and registrations are partially supported. The build fails, committed ERP provisioning is missing and many APIs only store records or set status.
No specialized ERP screens exist	Narrow to the 21 added unwired destinations. CRM, base HR directory and PM already have genuine routed screens.
34 pages and 25 modals remain	A proposal, not an accepted screen inventory. Page count is not workload; customer/employee flows and permission/error states need their own design.
Double entry means debit equals credit	Current code allows a 0.01 difference and lacks fiscal/account/concurrency controls. An approved rounding and posting policy is required.
PO reconciliation compares line items	The calling endpoint supplies an empty PO-item list, so same total and currency can produce a misleading Exact Match.
Month five AI engine and prediction	Excluded by current AI-free scope. Required face IN and deterministic dispatch still remain in scope.
200 metre radius and fixed SLA hours	Not adopted as company policy. Routing, GPS accuracy, SLA, OTP, authority and no-response decisions need sign-off.
Three quotes supplier portal and OCR integrations	Conditional proposals rather than automatically approved extra scope. Choose necessary channels and a manual verified fallback before pricing them.
Immediately generate and apply migration	Unsafe as an instruction for this checkout: repair build, choose the correct context, review schema and use an isolated database first.

Reuse and delivery decision

Current recommendation Conditional reuse candidate, not an approved replacement. Retain the NOIR platform/UI patterns and useful CRM/directory/project components as candidates. Treat the new ERP document and service code as unaccepted implementation requiring repair and business validation. Do not merge five or six entire systems or duplicate identities, customer/product masters, stock balances or financial ledgers.

There is no evidence here that borrowing code makes five months impossible, nor evidence that it makes full delivery feasible. The observed gaps make an unconditional full-SRS production promise unsupported. Compile defects may be quick repairs; the missing inventory, finance, attendance and mobile business engines are the larger delivery risk.

Capacity reality for three developers

Using an illustrative 21 working days per month, three full-time developers over five months provide 315 nominal developer-days. The supplied report already totals 277 to 342 developer-days across its 23 estimates. Removing its excluded AI panel leaves 257 to 318 days, without a demonstrated allowance for the broader HR v1.2 gaps, integrations, stabilization, QA, data migration, UAT, deployment or support.

For discussion only, 65 to 75 percent effective feature capacity would be about 205 to 236 days after meetings, review, rework and support. This is a sensitivity assumption, not measured team productivity or a new estimate. Calendar sequencing and three developers sharing specialist domains further limit simple division of effort by three.

First proof before choosing the permanent base

Use a proposed ten-working-day assessment timebox, not a promise to fix every issue in that time. Prove clean backend build and selected suites; reproducible fresh and upgrade SQL migrations; role/owner denial on selected APIs; one atomic supply/stock slice; one owned complaint-to-assigned-work slice; and the architecture fit for HR face/attendance without losing existing work. Record actual effort and remaining blockers. If these proofs fail, re-evaluate the foundation before adding modules.

Compare existing Cosmix HR/backend/mobile work separately before replacing it. Both inspected web project setups are .NET 10, so runtime version alone is not the barrier. Identity/session contracts, tenant versus branch/team scope, employee/customer/product keys, schema ownership, command interfaces and mobile sync are the integration work. Existing repository functionality was not certified by this review.

Rules for selecting any additional donor project

Record source URL, exact commit, original license/NOTICE, files copied, modifications, maintainers and security status. Require a compatible domain model and isolated business logic, not merely similar menu labels. Define one integration contract and migration owner; add positive, negative, retry and concurrency tests. Confirm redistribution and dependency-license obligations before commercial release. A root Apache license does not establish the provenance or licensing of every copied module.

The clone includes an Apache 2.0 license. Apache redistribution conditions include providing the license, retaining applicable notices, marking changes and retaining NOTICE attribution where applicable. Review the original license and donor provenance with the responsible company reviewer. Reference https://www.apache.org/licenses/LICENSE-2.0

Release hardening and pending evidence

Restore reported package advisory warnings for MailKit, Microsoft.OpenApi, ImageSharp and a test-side SSH.NET dependency. These are dependency metadata findings, not demonstrated reachable exploits. Review runtime reachability and supported patches with regression checks; do not certify security merely because a package comment says an older vulnerability was fixed.

Primary advisory references https://github.com/SixLabors/ImageSharp/security/advisories/GHSA-j3p4-wp97-rph4 and https://github.com/microsoft/OpenAPI.NET/security/advisories/GHSA-v5pm-xwqc-g5wc

Also resolve public bootstrap credentials/configuration, platform-admin boundaries and production secret injection; prove tenant and same-tenant owner controls. The Docker build stage contains no Node/npm installation while the web publish target invokes npm install/build. That deployment path was not exercised and needs correction/verification. The existing MCP server is an additional integration surface, not evidence of the excluded AI assistant; decide whether to disable it for the agreed release.

Code evidence NOIR.Web/appsettings.json line 5

Code evidence NOIR.Web/NOIR.Web.csproj line 29

Code evidence Dockerfile line 5

Management decisions that remain open

Confirm mandatory versus deferred modules and native/PWA delivery expectations; company/branch/team authority; store meanings and unit/valuation rules; AUX; approval/value/tax/rate versions; HR conflict resolutions, face provider and retention; geofence accuracy/radius; eligible-staff ranking and workload; complaint SLA/warranty/no-response closure; SMS/WhatsApp/mail accounts and provider terms; fiscal/posting/correction and payroll boundary; data migration and acceptance owners. Do not turn unresolved examples into hardcoded policy.

Evidence record and next review

Original specification C:/Users/Huma.Khan/Desktop/SOFTWARE REQUIREMENTS SPECIFICATION.docx

Current detailed AI-free specification C:/Users/Huma.Khan/Desktop/Cosmix_ERP_FSM_Detailed_SRS_v0.4.docx

HR specification C:/Users/Huma.Khan/Downloads/HR_Attendance_Employee_Mobile_SRS_v1.2.docx

Supplied claim report C:/Users/Huma.Khan/Downloads/Cosmix_Erp_software/Cosmix_new/HMS_ERP_FSM_Comprehensive_Development_Report.docx

Code anchors in this document link to the reviewed immutable GitHub commit and full source path. Their labels show the owning folder, filename and line. Detailed reviewer notes, extracted source records, source inventory, build logs and Domain TRX result are retained in Cosmix_new/analysis. They support the review but are not runtime/UAT acceptance records.

The next decision should be based on proof results and measured delivery of a selected end-to-end slice. A revised backlog and calendar can then be built for the three developers, with explicit acceptance owners and deferred scope. No application fixes, database migrations or repository merges were performed during this assessment.